

25PY101: Engineering Physics

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Summative preparation

QFET and QM

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Numerical Problem

Using the Fermi-Dirac distribution function, calculate the temperature at which there is a 2% probability of **not finding** an electron in an energy level that lies 2 eV below the Fermi level.

Q2. Graphical / Sketch-Based Question

Consider a particle confined in a one-dimensional box of width L .

- Sketch the **ground state wavefunction** for:
 - Infinite potential well
 - Finite potential well
- Repeat part (a) for the **second excited state wavefunction**.

Your sketches should clearly indicate:

- Boundary behavior of the wavefunction
- Nodes
- Penetration outside the well (if any)
- Relative shape of the wavefunctions

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: